For Loop (3 Questions)

1. Write a program to print the multiplication table of a given number using a for loop.

```
num = 5
for i in range(1,11):
    print(num,'*', i,"=",num*i)
```

```
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

2. Write a program to print all prime numbers between 1 and 100 using a for loop.

```
or num in range(2, 101):
    is_prime = True

    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break

    if is_prime:
        print(num, end=" ")
```

```
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97
```

3. Write a program to find the sum of all even numbers from 1 to n using a for loop.

```
  # Input from the user
n = int(input("Enter the value of n: "))

sum_even = 0

# Loop through numbers from 1 to n
for i in range(1, n + 1):
    if i % 2 == 0:
        sum_even += i

# Display the result
print("Sum of even numbers =", sum_even)
```

```
Enter the value of n:  5
Sum of even numbers = 6
```

2.While Loop (3 Questions)

1.Write a program to reverse a given number using a while loop.

```
  num = int(input("Enter a number: "))
rev = 0

while num > 0:
```

```
    digit = num % 10
    rev = rev * 10 + digit
    num = num // 10

print("Reversed number:", rev)
Enter a number:  456
Reversed number: 654
```

2.Write a program to check whether a given number is a palindrome using a while loop.

```
num = int(input("Enter a number: "))

original = num
reverse = 0

while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10

if original == reverse:
    print("Palindrome")
else:
    print("Not a palindrome")
Enter a number:  123
Not a palindrome
```

3. Write a program to calculate the sum of digits of a number using a while loop.

```
 n = int(input("Enter a number: "))

sum = 0
i = 1

while i <= n:
    sum = sum + i
    i = i + 1

print("Sum =", sum)
Enter a number:  5
Sum = 15
```

3.Functons(5 Que)

1. Write a function to calculate the factorial of a number.

```
def factorial(n):
    fact = 1

    for i in range(1, n + 1):
        fact = fact * i

    return fact


num = int(input("Enter a number: "))
print("Factorial =", factorial(num))
Enter a number:  5
Factorial = 120
```

2. Write a function that takes a list as input and returns the largest element.

```
def largest_element(lst):
    if not lst:
        raise ValueError("The list is empty.")
    return max(lst)

numbers = [10, 45, 3, 99, 27]
print(largest_element(numbers))
99
```

3. Write a function to check whether a given string is a palindrome.

```
def is_palindrome(s):

    cleaned = ''.join(char.lower() for char in s if char.isalnum())

    return cleaned == cleaned[::-1]



text = "A man, a plan, a canal: Panama"
print(is_palindrome(text))
True
```

4. What is a function in Python? Explain its advantages.

A function is a reusable a block of code that perfrom a specific task

Aviod code reptition

make code readable

improve moudalirtiy

Easier Debugging

5.Differentiate between user-defined functions and built-in functions with examples.

There are 71 bulit in functions in python bulit-in functions are predefined

User defined fuctions are created by the user

```
def greet():
    print("Hello python")
greet()
Hello python
```

4.OOPS (Theory – 9 Questions)

1. What is Object-Oriented Programming (OOP)?

OOP is a programming paradigm that organized software designed around objects rathter than function and logic

2. Explain the difference between Class and Object with an example.

Class is a blueprint or template using for creating objects

Object is a instance of a class

```
class                Object
@class method      instance method
```

cls parameter   self keyword

3. What are the four pillars of OOP? Explain each.

OOP is a programming paradigm that organized software designed around objects rathter than function and logic

Four

Inhertiance

Encapsulation

Polymorphism

Data Abstraction

1.Inhrtiance

A.Single/Simple-child class inherit from single parent class

B.Multiple- child class inherit from more than one patent

C.Mutilevel- In this inhertiance happpens in levels and setp by step inhertince

D.Hirechical - Mutpile child class inherit from single one parent

E.Hybrid - It is combination of inhertiance Hirechical + multiple multi-level + multiple

2. Encapsulation- Wrapping of Data

1.public x- access everywhere do not underscore prefix

2.protected _x - Defind using single underscore acccess inside + child class

3.private __x- Defind using double underscore acccseed inside the class\

3.polymorphism

one interface many forms

1.Method overloading- same method name but diferrent paramets- python does not supoort tradational overloading method

2.Method Overriding - Same method name same parameter

3.Using Inhertinace

4.Duck Typing - It Looks like a duck quack like a duck

5.Operator Overloading- Same operator behaves different output + numbers + string concatination + Merge List

4. Data Abstraction

Abstraction is a hiding internal implemenation of details and only showing only esstinal feaures to the user

Abstract class-Abstract class must be use inherit from ABC

Abstract method-Abstract method using @abstractmethod

import abc as ABC,@abstract method

we Cannot create object in abstract class

4. What is Inheritance? Explain its types.

Inhertiance used for code resuablity

A.Single/Simple-child class inherit from single parent class

B.Multiple- child class inherit from more than one patent

C.Mutilevel- In this inhertiance happpens in levels and -setp by step inhertince

D.Hirechical - Mutpile child class inherit from single one parent -one parent has common functionalties

E.Hybrid - It is combination of inhertiance Hirechical + multiple multi-level + multiple

5. What is Polymorphism? Explain Method Overriding and Method Overloading.

polymorphism means one mrthod one function differnt behaviour

1.Method overloading- same method name but diferrent paramets-

python does not supoort tradational overloading method

6.What is Encapsulation? What are its advantages?

Encapuslation is a wrapping of data

We Can Control for Class variables/method using access modifires

1.Public

2.protected

3.private

Advantages

1.Security

2.Data Hiding

3.Controlled Access

7.What is Abstraction? How is it implemented in Python?
Abstraction is a hiding internal implemenation of details and only showing only esstinal feaures to the user

Abstract Class

Abstract Method

Without Abstraction

1.Code Must be complex

2.Diffcult to matain

3.Higher changes of misuses

4.Uncessary featues to see the user

With Abstraction

1.Requried Features to be exopsed

2.Intrnal logic hidden

3.security

Data Abstract Class- Can Contain Absttract method(in-bulit), Act As a Blueprint

Abstraction Method- Has Declare Variable, No Implementation, Must be implemented in chlid class

we Cannot create object in abstract class

import abc from ABC,abstract method

8. What is the difference between Abstraction and Encapsulation?

Encapuslation Abstraction
Ddata Hiding | Implemntation Hiding

How to Data Is Protected | what functionalites are provided

Access Modifires | Abstract Class

9. Explain the use of the super() function in Python.

Super is in-bult python function use to the call mrthood or constructor of parent base class from child class

Super(). init()

super(). init(arguments)

Super(). method_name()

1.The Call Parent Class Constructor

2.Reuse The parent Class Code

3.Aviod Rewriting Code

Super is follow mro(Method Resolution Order) Decide to What parent method or call to constructor

1.Code resuablity

2.Code cleaner

3.Easier to Manange

It Is used in inhertiance

It Follow Mutpile In Inhertiance

Super(). init() intalizes the parent class before child class